

American Express Interview Experience

Role Offered: Technology Engineer Trainee

CTC: 20,65,393

Selection Procedure:

- Pre-Placement Talk (PPT)
- Online Assessment
- Online Interview

Online Assessment (09.08.2025):

The assessment consisted of three coding questions with a total duration of 1 hour 30 minutes.

Task 1:

Problem:

Given a string S of lowercase letters, return the minimum number of characters to remove such that the remaining characters form a string sorted in lexicographical order (for example, aabz is valid, but azb is not).

Input:

s = "banana"

Output:

3

Hint:

Find the length of the longest non-decreasing subsequence in the string, then subtract it from the total length to get the minimum deletions.

Task 2:

Problem:

You are given two strings S and T, both consisting of digits and representing two integers of equal length (possibly very large). You are allowed to swap digits at the same positions in the two numbers (i.e., swap the i-th digit of S with the i-th digit of T).

Your goal is to minimize the absolute difference between the two integers. Among all possible solutions, use the fewest swaps.

Input:

S = "29162", T = "10524"

Output:

2

Hint:

Compare digit by digit and identify the positions where swapping reduces the most significant differences first.

Task 3:**Problem:**

You are given a rectangular board with N rows and M columns. Each cell contains a single digit (0–9). You need to find a path of exactly four distinct neighboring cells. Two cells are neighbors if they share a common side (up, down, left, or right). By visiting these four cells in order, you form a 4-digit integer. Return the largest possible integer that can be formed.

Input:

Board =

[

[2, 9, 1],

[4, 0, 7],

[3, 6, 5]

]

Output:

9765

Hint:

Use DFS or BFS from every cell (avoiding revisits) to explore paths of length 4, and track the maximum number formed.

Online Interview- Round 1 (13.08.2025):

The interview was conducted on Webex. Candidates were asked to join based on their allocated slots. My panel consisted of 2 interviewers.

Interview Flow:

1. Introduction – The interview began with a short introduction about myself.
2. Programming Languages – I was asked to rate my proficiency in the programming languages mentioned in my resume. Since I rated C++ higher, most questions were from C++.
3. C++ Questions – I was asked to code multiple inheritance with specific class names and methods provided by the interviewer. They also asked

about accessing parent methods using the child class and questions on pointers in C++.

4. SQL Question – I was asked to write a query to find the second maximum salary from an Employee table.
5. OOP Concepts – I was asked about abstraction in object-oriented programming.
6. Project Discussion – They asked questions about my project's backend code and how the frontend was connected with the backend.
7. Situation-Based Question – I was asked, “How will you manage the workload in a high-pressure situation?”

Round 2:

This time I was assigned a panel consisting of 1 member.

The round started with a short self-introduction. After that, the interviewer focused mainly on Java, DBMS concepts, DSA, and a few HR-type questions.

Topics covered in Java:

- Difference between Interface and Abstract Class – I was asked to explain with examples.
- Final keyword – explained its usage with variables, methods, and classes.
- Is multiple inheritance possible in Java? – explained with reasoning and how interfaces solve this problem.
- Java 8 Features – explained features like Lambda Expressions, Functional Interfaces, Streams API, and Default Methods in Interfaces.

DBMS Questions:

- Concepts of Normalization – explained step by step with example tables.
- Left Join – explained with example queries and how it differs from Inner Join.
- Discussed practical use cases of joins in real-world scenarios.

DSA Question:

- Find the second largest element in an array.
- I was asked to dry run all the iterations and clearly explain what happens in each step.
- The interviewer also questioned how the logic would handle duplicate elements and edge cases like an array with all identical numbers.

Company-related Questions:

- Why American Express? – I explained about the company culture, values, and focus on technology and customer-centric services.
- What do you know about American Express? – I spoke about its history, role as a financial services giant, its credit card business, and reputation as one of the Fortune 500 companies.

Closing:

Finally, I was asked if I had any questions.

I asked, “How does a day look like for a software engineer at American Express?”

The interviewer explained about work culture, daily tasks, collaboration with teams, and opportunities for growth.

SANTHOSH S

MIT CT'26

Santhoshsiva2909@gmail.com